A Protocol for Validating Pairing Assumptions in Seed-Matched Evaluations of Black-Box Game-Playing Agents

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(Dated: August 24, 2026)

When two game-playing agents are evaluated under the same recorded seed, the shared value names a matched schedule; it does not by itself establish repeatable execution or equal random quantities for the same semantic events. We present an executable, fail-closed protocol that validates artifact identity, boundary seeds, row-level schedules, identical-arm trace projections across execution contexts, stochastic sources, cross-arm event evidence, and statistical claim admission, downgrading or suppressing paired wording whenever required evidence is absent. A standalone synthetic suite reproduces seed-conversion, stateful draw-shift, timing, and process-state failures, demonstrates event-keyed repair for a declared ontology, and rejects schema or manifest tampering. In a restricted-engine case study, a deterministic preflight showed zero required mismatches across 3,000 executions. A fixed timed-search battery disagreed on the recorded trace projection in 99 of 200 seed-condition clusters. One apparently favorable historical comparison failed a repeated-control repeatability audit and was therefore suppressed, while a later comparison acquired under a narrower frozen rule was admitted even though its policy contrast remained unresolved; admission tracked evidence rather than favorability. Because the restricted engine exposes no semantic event identifiers, the case cannot establish event-aligned coupling.

I. INTRODUCTION

Reusing random inputs can sharpen a comparison when two stochastic systems respond similarly to the same sources of variation. Common-random-number (CRN) theory makes that gain conditional on the construction and on properties such as system structure and event timing, rather than on the reuse of an integer alone [1]. Streams and substreams are established ways to organize synchronized simulation and independent replications [2]. A recent paired-seed preprint likewise studies the conditional precision gain when seed-level outcomes are favorably correlated [3]. These results motivate careful pairing; they do not turn a seed column into evidence of a particular coupling.

The gap matters for black-box game-agent evaluations. A researcher may be able to hash a restricted simulator, call a seeded entry point, freeze an opponent schedule, and record public actions, yet remain unable to inspect the simulator's generator or assign stable identifiers to random events. Separate executions can also encounter clock-bounded search, process-local state, or different queue histories. Under those conditions, the same recorded seed can name a matched design without establishing repeatable execution or semantic alignment across arms.

This article addresses one question: when two agents are evaluated under the same recorded seed, what evidence is required before the result can legitimately be analyzed as a paired comparison? We integrate four distinct evidence layers—schedule matching, within-arm execution repeatability, cross-arm event alignment, and statistical claim admission—into an executable validation protocol for restricted black-box settings. The protocol records artifact and seed-boundary identity, audits every scheduled row, compares a declared trace projection in A/A and repeat/worker tests, inventories stochastic sources, and links gate evidence to bounded wording. It fails closed: a later point estimate or test cannot repair a missing gate. For the completed case, frozen binary acquisition and suppression rules are kept distinct from a post-acquisition conservative claim taxonomy; future users must freeze that taxonomy before collecting outcomes.

Prior work supplies conditional CRN theory, structured random streams, trace preservation, deterministic testing, and event-keyed randomness. This work integrates those components into an executable protocol for validating the pairing assumptions behind statistical comparisons in black-box stochastic agent evaluation and maps the evidence obtained to the paired claim that may be admitted. The restricted game environment is a case study, not the title-level contribution, and no competitive-strength claim is made.

II. RELATED WORK AND CONTRIBUTION BOUNDARY

CRN conditions, stream design, paired-seed evaluation, counter-based random access, and event-keyed randomness already cover the central stochastic-computation ingredients [1–5]. The Sharma and Buffalo papers are arXiv preprints, not verified peer-reviewed publications. Sharma treats a seed as the pairing unit when studying conditional precision; this paper tests what a restricted implementation must show before inheriting that assumption. Buffalo, Pearson, and Klein establish the state-

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ful draw-shift problem and event-keyed white-box repair. This paper does not claim either idea as new and cannot validate the restricted engine's unobserved event ontology.

Three close 2026 works strengthen the boundary. Rollout Cards preserves rollouts, declared views, reporting rules, and dropped-record manifests [6]. The trace-assurance framework of Paduraru, Bouruc, and Stefanescu uses Message–Action Traces for contracts, replay, fault injection, localization, and governance [7]. AEVAL turns agent-workflow changes into deterministic evaluation-contract tests and preserves first-attempt evidence [8].

Two further 2026 evaluation-auditing works are closest in spirit. Protocol cards disclose the protocol conditions behind coding-agent scores [9]. A paired noise-floor protocol bounds same-configuration paired gaps across repeated seeds and retracts contrasts that fail to replicate [10]. Neither validates seed-to-execution coupling through within-arm trace repetition across execution contexts, and neither maps failed pairing evidence to automatic suppression of a statistical claim. Together with the three works above, they improve evidence preservation, protocol disclosure, and reporting honesty; none of the five validates same-seed stochastic coupling. Recent empirical studies of randomness and repeated-trial inconsistency in agent evaluation further establish that run-to-run variation itself is not an unoccupied topic [11, 12]. The broader empirical literature distinguishes reproducible code from reproducible findings, emphasizes explicit units and sources of variation, and recommends uncertainty-aware reporting across heterogeneous tasks [13–16]. Simulation verification and validation distinguish implementation checks from operational validity [17]. Metamorphic testing has been applied both to scientific software without a convenient single-run oracle and to simulation verification and validation [18, 19]. A/A experiments are also an established diagnostic in randomized online experimentation [20]; the identical-artifact trace repetition used here adapts that diagnostic idea, rather than importing its sampled-user inference. Wall-clock allocation is already an algorithmic design concern for Monte Carlo tree search [21]. That literature supports auditing time-bounded search but does not identify the cause of a particular observed divergence.

The boundary in Table I is therefore deliberate. The protocol does not replace simulator validation, prove favorable seed-level covariance, or guarantee external generalization. It supplies a black-box workflow that turns prior component methods into an explicit decision about which paired claim, if any, the recorded evidence permits.

Although the case study is a game engine, the evidence interfaces are not game-specific. Simulation methodology has long distinguished implementation verification from operational validity [17], and empirical work across machine learning emphasizes explicit units and sources of variation [13, 14]. Monte Carlo studies, agent-based models, stochastic numerical experiments, scientific-machine-learning benchmarks, and digital-twin simulators all face the same adoption question: does a shared seed or matched schedule license paired analysis of two configurations? A setting that exposes a seeded entry point, hashes its artifacts, and records per-run outcomes can apply the identity, schedule, and repeatability stages directly; alignment stages apply wherever stable event identifiers exist. The protocol thereby offers computational and adjacent sciences a concrete path, while prevalence of these failure modes in any new domain remains unmeasured until audited.

III. DEFINITIONS

A. Schedule matching

A schedule is the prospectively declared set of evaluation rows. Two arm rows are schedule matched when all pairing variables required by the protocol agree: the value passed at the observable engine boundary, opponent, actual order, physical seat, environment, decision limit, and other fixed configuration. This is a property of design records. It says that a matched comparison was attempted; it does not say that either execution is repeatable.

B. Execution repeatability

Execution repeatability is a within-arm property. For the same artifact, seed condition, and declared execution context, the compared execution record must agree across fresh runs. Here the strongest trace-bearing record is a canonical sequence of public-observation hashes, acting side, selected action, and terminal record. The implementation compares the SHA-256 digest and byte count of that complete recorded projection. It does not retain raw public observations, opaque search snapshots, hidden simulator state, or semantic random-event identifiers. Thus "complete recorded trace projection" always means complete only within this declared projection.

The frozen case-study stress endpoint compared trace digests. Adding byte-count equality is a post-acquisition integrity extension: the retained stress battery has 99/200 disagreements by digest alone and the same 99/200 by the extended digest-or-byte rule. The byte-count-only toy case below tests the current definition; it is not evidence that this extension was prospectively specified.

Trace agreement is stronger than outcome agreement because different action sequences can end with the same winner. Conversely, a digest is not a proof that omitted state agreed. A single required mismatch rejects exact repeatability on the exercised schedule; a pass remains scoped to the tested artifacts, projection, and contexts.

For a stress test with N seed-condition clusters, let $H_{ir} = (D_{ir}, B_{ir})$ be the ordered pair comprising the SHA-

TABLE I. Relationship to prior methods. The present contribution is their operational integration for black-box agent evaluation, not priority over the component ideas.

Prior area	Established contribution	Role here
Common random numbers	Conditions for covariance and variance reduction; synchronization depends on model structure and event timing [1]	Motivates separating a matched integer from an evidenced coupling.
Streams and substreams	Organized independent replications and synchronized streams [2]	White-box design option; stream labels alone do not prove semantic assignment.
Paired-seed evaluation	Precision can improve when seed-level outcomes are favorably correlated [3]	Supports pairing when its implemented relationship is validated.
Event-keyed and counter-based randomness	Stable event keys prevent draw shifts within a declared event ontology; counter-based generators provide random access [4, 5]	Implemented only in the synthetic repair; unavailable inside the restricted engine.
Agent evidence records and deterministic workflow tests	Rollout records preserve views, reporting rules, and drops; evaluation contracts turn workflow changes into auditable tests [6, 8]	Complementary evidence preservation and testing; neither validates paired stochastic coupling or admits a statistical claim class.
Trace-contract assurance	Message–Action Traces support contracts, replay, perturbation, localization, and governance [7]	Closest trace framework; it does not distinguish identical-arm repeatability from cross-arm event alignment or suppress paired inference.
Reproducibility and RL evaluation	Variation, units, estimation, and transparent reporting are essential [13–16]	Supplies the broader empirical-design boundary.
Software and simulation testing	A/A, metamorphic relations, and verification/validation test implementations and models [17–20]	Informs trace invariants and fail-closed conformance checks.
Work versus time budgets	Search time allocation is established prior art [21]	Justifies auditing clock-bounded search without claiming its invention.

256 digest D_{ir} and byte count B_{ir} of the required trace projection for cluster i under execution profile r. The trace-disagreement proportion is

$$\widehat{q} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(|\{H_{ir} : r \in \mathcal{R}\}| > 1)\,. \tag{1}$$

Here $\mathcal{R}$ is the complete prescribed set of profiles, the indicator is one when at least two required digest–byte-count pairs differ, and $\widehat{q}$ is a unitless proportion of seed conditions. A missing required profile invalidates the gate rather than being omitted from the set or denominator. In plain language, the equation counts seed conditions whose repeated recorded projections disagree. For example, if two of five complete clusters disagree—including one in which the digest agrees but the byte count does not—$\widehat{q} = 2/5 = 0.40$. The executable example and fail-closed missing-profile check are recorded in `tests/test_equations.py`.

C. Semantic event alignment

Semantic event alignment is a cross-arm property. After two policies branch, every shared modeled exogenous event in the declared ontology must retain a stable identifier and receive the same random quantity. Logged event/value pairs, event-keyed counter-based randomness, or a validated assignment of event-specific streams can provide this evidence [2, 5]. Exact within-arm repetition cannot establish it: identical policies ordinarily follow the same path and consume the same stateful sequence, while different policies can shift later draws.

An event-alignment pass is conditional on the ontology, marginal random distributions, dependence structure, and logged event classes. The restricted engine lacks these event identifiers, so its evaluations cannot establish this property.

D. Descriptive pairing, inferential pairing, and admission

A schedule-indexed descriptive pair consists of two observed outcomes carrying the same frozen opponent, order, seat, boundary seed, and other declared row fields. It permits the algebraic within-row difference used below, but it does not by itself supply a probability distribution for that difference. Inferential pairing is an additional design claim: a randomized assignment, probability-sampling mechanism, or defended joint stochastic model must justify the reference distribution and the independence or exchangeability of analysis units. Semantic event alignment is different again; it asks whether modeled exogenous events receive the same quantities across arms. Neither descriptive pairing nor ordinary matched-pair inference automatically establishes that stronger property.

Statistical admission maps the evidence achieved to an estimand, analysis unit, descriptive or inferential procedure, target, and exact wording. Schedule evidence can authorize only a matched-schedule description. The completed case study's frozen rule gated acquisition and suppression but, on a pass, specified 95% paired-resampling intervals and secondary McNemar inference. It did not contain the later named Level 6 descriptive taxonomy. After acquisition, a conservative methods audit judged inferential wording unsupported without a randomized assignment, probability sample, or defended joint stochastic model. This article therefore reports strictly less: fixed-battery differences and the unchanged prespecified resampling quantiles, relabeled as descriptive empirical sensitivity. That post-acquisition restriction is not evidence of prospective claim-map validation. Required gate failures still suppress comparisons under the frozen rule, and event-aligned wording still requires semantic event evidence.

Figure 1 summarizes the non-implications among these definitions.

IV. PAIRING-ASSUMPTION VALIDATION PROTOCOL

The protocol solves an admission problem, not a policy-selection problem. Its analysis unit is defined at each stage: artifacts at identity, arm-paired rows at schedule parity, arm–seed conditions at preflight, seed-condition clusters containing all execution profiles at stress testing, and paired game units within frozen opponent-by-order strata at factorial estimation. Opponent and worker profiles are fixed execution contexts, not random samples from a larger population. The case-study seeds are also a prospectively fixed engineering battery rather than a probability sample. Consequently, its Level 6 factorial output is descriptive; the protocol would require a separately declared sampling, assignment, or stochastic model before admitting paired population inference.

A. Stages 1–3: identity, boundary seed, and schedule

Stage 1 hashes the simulator, policies, opponents, protocol, evaluator, and configuration. The problem it solves is artifact ambiguity: a result without identified bytes cannot be attributed to the declared system. A hash match does not validate behavior. Missing or drifting artifacts invalidate the affected run.

Stage 2 records the scheduled host-language seed and the exact integer passed at the observable engine boundary. This is needed because an adapter may narrow, reduce, or otherwise convert a value. The full schedule is converted before execution and checked for unintended collisions and overlaps. The engine's internal handling remains unobserved; the protocol never calls the boundary value an internally consumed seed.

Stage 3 checks equality and completeness for every required schedule field. The output is a row-level parity report and schedule fingerprint. A missing cell, unequal field, error, or silent deletion invalidates the comparison. A historical data set that lacks fields required by the current protocol can support agreement only for its recorded projection, not a retrospective pass of the full stage.

B. Stages 4–6: trace repetition and source audit

Stage 4 is within-artifact fresh-process repeatability: byte-identical arms are re-executed in separate fresh processes on the same schedule rows and their records are compared in increasing strength—outcome, errors, decision count, and the digest plus byte count of the recorded trace projection. This adaptation of the A/A diagnostic detects implementation or execution differences that an outcome check can miss. Its assumptions are that the serialization is canonical, the digest comparison is implemented correctly, and all fields relevant to the claim are declared. The verifier checks the declared artifact hashes, the shared schedule-row identifiers, and the two fresh-process records per arm; it cannot authenticate launch provenance, so which bytes actually ran in which process remains a caller assertion anchored outside the file-bound verifier. Stage 4 does not establish cross-arm coupling.

Stage 5 prespecifies execution-context variation: fresh serial repeats, worker counts, enqueue orders, and pool lifecycles. The stage-4 fresh-process profiles are a subset of the stage-5 required profile set, so the two gates are nested rather than independent; stage 5 adds the worker and reversed-enqueue contexts. Repeats within one seed condition share the same planned inputs and are not independent observations. Worker profiles test execution contexts; they are not sampled hardware replicates. The output is a cluster-level trace-parity report with machine-checkable profile sets recorded in each gate certificate. Any required mismatch rejects exact parity for at least that exercised seed condition.

The implemented Stage 6 is a bounded Python AST and source-pattern scan of package trees. It inventories explicit random interfaces, clock or deadline patterns, concurrency APIs, module-level state, and native pointer or handle patterns. A static hit identifies a mechanism for dynamic testing but does not show that the branch executed or caused a mismatch; a zero-hit scan does not prove determinism. The scanner does not inspect binary-engine internals or classify hardware and external state. The stage is invalid if the inventory or its scope is missing, and it cannot identify a unique cause. Every audit resolves to exactly one of five outcomes. CONTROLLED: a mechanism was found and its influence bounded or disabled with recorded evidence. RESIDUAL: a mechanism remains possible and is not shown estimand-changing, so claims are downgraded. ESTIMAND-CHANGING: a mechanism directly changes the execution relation needed by the comparison, so paired mechanistic wording is suppressed. UNAVAILABLE: the source sits behind a black-box bound-

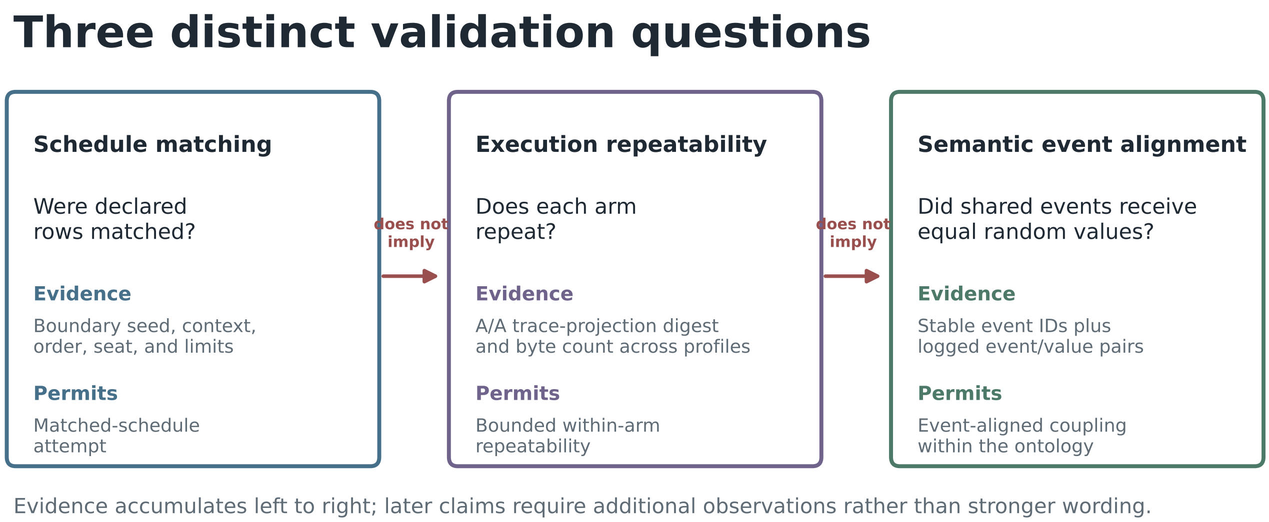

FIG. 1. Schedule matching, within-arm execution repeatability, and cross-arm semantic event alignment answer different questions and require different evidence. Each claim is limited to the declared schedule, trace projection, execution contexts, and event ontology. Source data are in `source_data/figure_1_distinctions.json`.

TABLE II. Protocol stages, required evidence, strongest permitted statement, and failure action. Every pass is local to the tested artifacts, schedule, trace schema, and execution contexts.

Stage	Evidence	Permitted statement	Failure action
1 Artifact identity	Hashes for engine, policies, opponents, protocol, code, and configuration	The evaluated bytes are identified.	Quarantine missing or drifting artifacts.
2 Seed namespace	Scheduled seed, exact boundary value, conversion rule, and collision audit	The claimed boundary seed was supplied and units remain distinct.	Correct the adapter or schedule and rerun.
3 Schedule parity	Row equality for opponent, order, seat, seed, environment, and limits	A complete matched schedule was attempted.	Reject unequal or incomplete pairs.
4 Identical-arm parity	Separate A/A executions compared through the digest and byte count of the recorded trace projection	The compared fields repeat on exercised A/A trajectories.	Suppress the affected paired comparison.
5 Repeat/worker parity	Fresh repeats, worker counts, enqueue orders, and process lifecycles	Within-arm traces repeat in tested contexts.	Freeze a passing context or model run variation.
6 Source audit	Bounded Python source-pattern scan; binary internals remain unassessed	Candidate mechanisms and the uninspected boundary are recorded.	Control, dynamically test, or narrow the estimand and wording.
7 Event alignment	Stable event identifiers and equal values for shared exogenous events	Event-aligned coupling for the logged ontology.	Retain at most bounded descriptive seed-indexed wording.
8 Statistical admission	For future use, a prospectively frozen map from gates to estimand, unit, procedure, target, and wording; for this completed case, the frozen experimental plan and later generic taxonomy are kept distinct.	The frozen case plan prescribed paired-resampling intervals and secondary McNemar inference; this article's post-acquisition taxonomy reports only a Level 6 fixed-battery description and empirical sensitivity.	Apply the authoritative frozen rule; a later conservative taxonomy may narrow reporting but cannot be described as prospective.

ary, blocking stronger source-level claims. UNRESOLVED: a hit exists without sufficient evidence, and no automatic pass exists. Each outcome carries a required remediation, and the mapping is generated from machine-readable rules shipped with the companion rather than from ad hoc judgment.

C. Stages 7–8: alignment and claim map

Stage 7 compares stable identifiers and random values for shared events after policy branching. The output names the covered event classes and any misalignment. Because the case-study engine exposes neither event keys nor independent event streams, the stage is unavailable there. This blocks event-aligned, counterfactual, and full-CRN wording but does not erase the lower-level evidence.

Stage 8 has two provenance layers. The frozen experiment-specific rule applied binary acquisition and suppression actions and, on a pass, planned seed-matched intervals and secondary McNemar inference. The Level 6 wording shown in Fig. 2 is a post-acquisition conservative taxonomy. For the completed case it permits only a fixed-battery descriptive contrast and empirical reweighting; for a future study it must be frozen before acquisition. Neither layer can use a favorable result to waive an earlier failure. The analyzer also rejects nonfinite results, schema or commit changes, sample-size drift, missing strata, and requests for suppressed output.

D. Executable admission map and formal properties

The computational companion expresses generic claim classes, evidence states, decision rules, and the file-bound record schema in four machine-readable JSON files and generates its human decision table from those files. This generic executable taxonomy was created after acquisition as a conservative restriction on what the article reports. It did not exist at the protocol commit, cannot be credited as prospective validation for the completed case, and deliberately authorizes less than the frozen plan's inferential outputs. The original binary failure actions remain authoritative historical constraints.

The evidence-bound `admit` path checks a closed support tree, hashes declared regular files, checks trace byte counts, and derives gate states from structurally validated records. An explicitly trusted-state classifier is a separate interface for formal conformance checks whose caller supplies those states as trusted inputs. The generic rule evaluation receives only the seven gate states and declared trace projection. Observed outcomes, treatment estimates, reweighting quantiles, p values, and result favorability are outside its input projection. This establishes declared-byte binding and cross-profile record checks only. The generic verifier does not parse engine-specific trace semantics or prove artifact roles or execution-profile provenance; those remain caller assertions and require an independently controlled acquisition manifest or equivalent external trust anchor. Machine verification of arbitrary external evidence therefore extends fully through the lower stages only. Successful Level 6 and Level 7 demonstrations are currently available through the synthetic conformance suite's declared adapters and the explicitly trusted-state interface; reusable file-bound evidence schemas for completed source audits and event-alignment records remain future work, and the bundled example marks both upper gates conservatively rather than manufacturing stronger evidence.

a. Admission-safety proposition. For any structurally valid evidence record, the implemented admission map is result independent, prerequisite monotone, failure dominant, projection scoped, statelessly deterministic, and fail closed for unknown states. Strict claim ordering counts as a seventh property. Missing, malformed, contradictory, or unrecognized evidence can never strengthen a claim. Each property is backed by checked property tests that replace all result values, weaken every prerequisite, inject extreme favorable results after failures, remove trace-projection scope, repeat identical evaluations, and enumerate every combination of the six evidence states across seven gates. Protocol, schema, and claim-class tampering are rejected.

In ordinary language: changing a favorable outcome payload to an unfavorable one leaves the gate decision unchanged (result independence). Removing evidence or turning an earlier pass into a failure can never strengthen the wording, regardless of a later estimate (monotonicity and failure dominance). Repeatability is confined to the named recorded fields (projection scoping). The same evidence record returns the same answer on every evaluation; no mathematical idempotence claim is made. Missing, malformed, contradictory, or unrecognized states map to suppression. For example, if Levels 1–4 pass but the repeat/worker gate fails, a carried result of either $+20$ or -20 percentage points yields the same schedule-matched downgrade; deleting artifact evidence weakens it further rather than rescuing the claim.

An engine-independent internal-consistency example passes declared-file identity, seed, schedule, and identical-arm record checks but fails the required repeat/worker gate; it is not evidence that external executions occurred. Running `python -m pevl_bench admit examples/example_evidence.json` returns the `schedule_matched` class; `explain` names the blocking gate, forbids a paired treatment-effect claim, and recommends reacquisition across the required fresh-process profiles. The companion also exposes `generate`, `verify`, and `report` commands and ships the generated Stage-6 decision table described above.

E. Seed-indexed descriptive contrast and reweighting

For binary outcome Y (win $=1$, nonwin $=0$), the paired difference for unit i is

$$d_i = Y_i^{(I)} - Y_i^{(C)}. \tag{2}$$

The superscripts denote intervention and control; each Y and d_i is unitless, with $d_i \in \{-1, 0, 1\}$. The fixed-battery

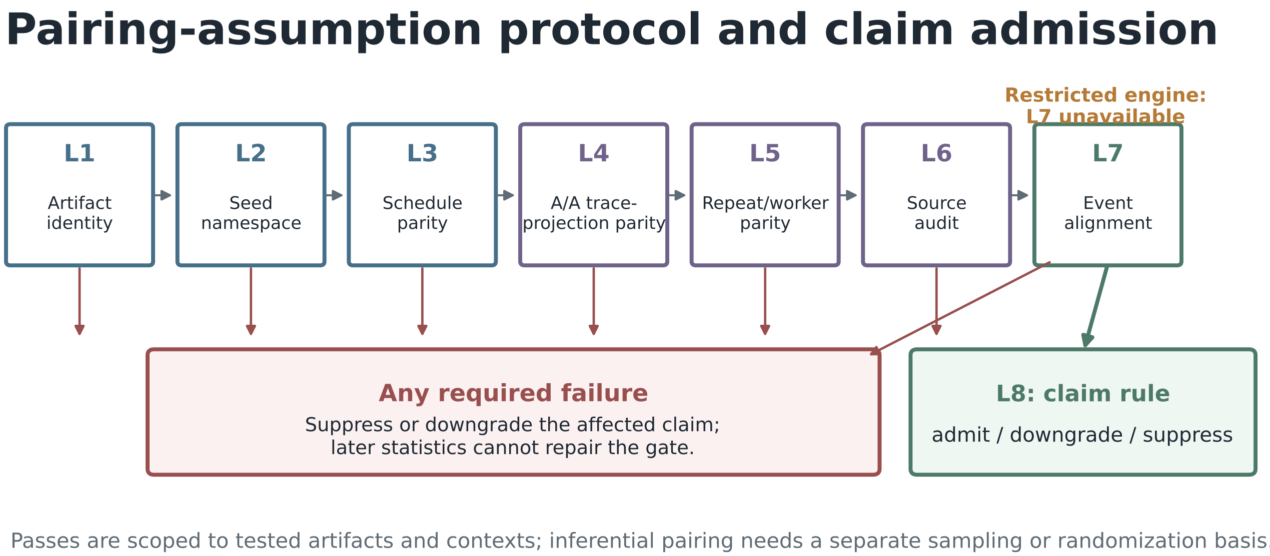

FIG. 2. Validation and admission flow. Levels 1–7 accumulate scoped evidence. A frozen required failure retains its suppression action; Level 8 cannot repair it. The colored claim classes are a later conservative reporting taxonomy for this completed case and a template to freeze before a future acquisition. Level 6 does not itself supply a sampling or randomization basis for inferential pairing. Source data are in `source_data/figure_2_admission_flow.json`.

mean is reported as a percentage-point win-rate difference. In ordinary language, d_i records which arm won on one schedule-indexed unit. If intervention wins and control does not, $d_i = 1 - 0 = 1$. This algebraic identity is independently tested in `tests/test_equations.py`. It does not itself supply an inferential reference distribution or show that the paired outcomes are semantically event aligned.

For K frozen strata with n_k paired units, empirical reweighting replicate b is

$$\bar{d}^{*(b)} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{n_k}\sum_{j=1}^{n_k} d_{k,I_{bkj}}, \qquad I_{bkj} \sim \mathrm{Uniform}\{1,\ldots,n_k\}. \tag{3}$$

The index I_{bkj} reweights whole schedule-indexed paired-unit identifiers within stratum k by sampling their indices with replacement; arms are never reweighted separately. Because strata may contain unequal unit counts, the outer $1/K$ makes each replicate an equally weighted mean of stratum means rather than an unqualified battery mean over all units; with equal n_k the two coincide. Conditional on each fixed empirical stratum, the I_{bkj} are independent draws for this reweighting calculation. The result is a unitless mean, reported in percentage points. With one stratum, differences $[1, 0, -1]$, and sampled indices $[1, 1, 2]$, the replicate is $(1 + 1 + 0)/3 = 2/3$. The equation assumes the frozen within-stratum empirical distribution is the object being resampled. Here that operation is interpreted only as empirical reweighting. Its 2.5th and 97.5th percentiles are a descriptive sensitivity summary for the realized battery, not a confidence interval, hypothesis test, or population-effect statement over new opponents, seeds, training runs, or execution contexts [22].

F. AI-assisted research methods

OpenAI Codex, using a GPT-5-family model for which the exact deployed snapshot was not exposed, provided substantive research assistance under human direction. Recorded tasks included protocol reasoning, literature synthesis, code generation and debugging, simulation orchestration, statistical-analysis code, deterministic visualization, and adversarial auditing. The human author set the scientific question, froze the acquisition protocol, retained authority over experiments and claims, and must approve the final paper. Verification included artifact hashes, strict schemas, row-level reaggregation, independent equation tests, targeted software tests, comparison against primary literature, a one-command reproduction run, compilation, and page-image inspection. The machine-readable activity record is `supplement/AI_USE_LOG.csv`. No generative-image system was used; all figures are deterministic plots produced by the checked-in artifact builder.

V. SELF-CONTAINED SYNTHETIC CONFORMANCE SUITE

The standalone standard-library suite supplies an inspectable computational test oracle without the restricted game engine. Five fixed modes exercise distinct non-implications: a clean deterministic mode, a stateful draw shift, an injected clock-budget mode, an injected process-state mode, and a seed conversion mode. Injected profiles make the fixtures reproducible; they are demonstrations of mechanisms, not claims that every external simulator contains them.

The seed-conversion fixture repeats within arm while failing namespace integrity. The clock and process fixtures fail trace repetition and execution- context checks. The stateful draw-shift fixture repeats exactly within each arm but aligns only one of five shared events after one arm consumes an extra draw. Deriving random quantities from the base seed and semantic event key aligns all five logged shared events. This repair is established only for the fixture's declared event ontology and source distribution.

The suite writes complete JSON evidence, a level-by-mode CSV, a JSON Schema, and a SHA-256 manifest. Within the computational companion, generation and verification are exposed as `python -m pevl_bench generate` and `python -m pevl_bench verify`. Checked tests validate the five fixed mode decisions, event-keyed repair, schema rules, and manifest-tamper rejection.

VI. RESTRICTED GAME-AGENT CASE STUDY

The case study uses a seeded tournament-engine binary, four fixed policy packages, and third-party opponent packages that cannot be redistributed under the currently established rights. Policies are described only as control, representation intervention, training intervention, and combined intervention. Opponents are grouped as deterministic or timed-search execution contexts. The scientific target is the validation protocol; policy quality and game strategy are outside scope.

The bounded source audit verified frozen digests for all 13 inventoried artifacts and applied the Python source-pattern scan to 11 package trees. The 2 engine binaries were hash checked but not source assessed. These records identify candidate mechanisms and an explicit uninspected boundary; they do not prove determinism or a causal mechanism.

The historical study scheduled seven opponents, two actual orders, and 200 seed conditions per opponent–order cell. Candidate and control rows recorded the same requested seed, opponent, order, and seat, but ran as separate process-pool tasks. The historical files do not contain the complete field set required for a current Stage-3 pass and did not capture the later trace projection. The strongest retrospective check is therefore equality of an available-record projection across three separately executed control arms.

That projection disagreed on 210/2,800 units when it included win and draw fields and on 458/2,800 units after decision count was added. Both mismatch sets were confined to two timed-search opponent packages. That audit rule was frozen only for retrospective reanalysis after the historical outcomes already existed. Its rule consumes only repeated control-arm records, so it is independent of candidate-effect magnitude, direction, or favorability, but it is not outcome blind in a literal sense and the exercise is not evidence of prospective blinding or pre-acquisition specification. An apparently favorable estimate failed that retrospective control gate, so the rule suppressed the historical paired comparison. The mismatches are evidence of failed repeatability for those records, not proof that wall-clock timing alone caused every divergence. Removing the two opponents after inspecting the failure would have changed the target population and was not used as a rescue analysis.

VII. PROSPECTIVE VALIDATION RESULTS

The experiment-specific protocol content first appears at commit 803257f1, and the retained result artifacts appear in descendant commits. Repository history establishes only that Git ordering; it cannot independently prove when a human inspected uncommitted files. Property tests also establish functional result invariance: replacing an outcome payload while holding gate evidence fixed does not change admission. They do not establish human blinding. Artifact hashes, the boundary seed conversion, schedule rows, trace schema, execution profiles, sample sizes, resampling seeds, and binary failure actions are recorded in that protocol commit. The same frozen plan prescribed 95% paired-resampling intervals and secondary exact McNemar inference. The frozen protocol already enumerated the full eight-stage evidence ladder with binary gate consequences; what was added after acquisition is narrower: the generic completed-case claim-class taxonomy, the descriptive-only reinterpretation of those intervals as empirical reweighting, and a digest-plus-byte integrity extension. None of these later additions is attributed to that commit, and the completed case does not prospectively validate them; future users should freeze them before collecting outcomes. The prospective boundary-seed audit found all scheduled values in range and distinct, with no overlap with the converted historical namespace.

A. Deterministic preflight

The preflight covered four arms and five deterministic contexts. These five contexts were a newly frozen, audit-informed target selected for the new protocol and recorded in the experiment-specific commit that precedes

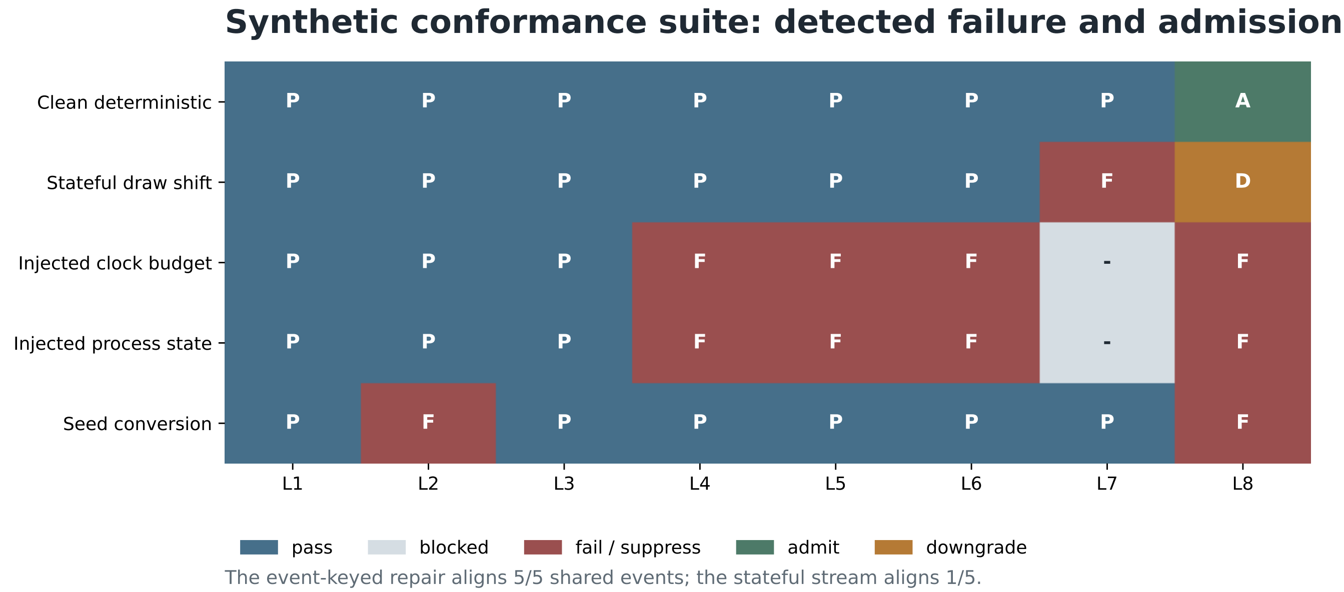

FIG. 3. Synthetic failure-detection and admission matrix. P denotes pass, F fail, A admit, D downgrade, and a dash a blocked gate. The suite demonstrates known logical relations; it does not estimate failure prevalence in external systems. Source data are in `source_data/figure_3_synthetic_matrix.csv`.

TABLE III. Prospective validation results. Trace-projection digest and byte-count fields were captured for preflight and stress executions but not for factorial outcome rows.

Stage	Units/clusters	Executions	Trace-record disagreement	Outcome disagreement	Admission
Deterministic preflight	1,000	3,000	0	0	Qualify acquisition for the newly frozen five-context scope; later-seed transfer remains an assumption.
Timed-search stress	200	800	99	47	Reject exact repeatability for at least one exercised seed condition.
Factorial repeated control	2,000 per cell	12,000 games	—	0	Frozen plan admitted its paired analysis; the later conservative taxonomy reports a fixed-battery description plus reweighting sensitivity and no Level-7 claim.

the retained artifacts; they were not a rescue subset or a retrospective redefinition of the historical seven-opponent target. Each arm–opponent job contained both actual orders, 25 seeds per order, two fresh serial profiles, and one eight-worker profile. Across 1,000 arm–seed-condition units and 3,000 execution trajectories, the frozen trace-digest, terminal-outcome, error, and decision-count checks had 0 mismatches. A later audit of the recorded byte counts also found zero and did not alter the decision. This passed the frozen digest-based repeatability gate and admitted factorial acquisition. The preflight and factorial used different frozen seed ranges. Treating this qualification result as applicable to the later factorial is therefore a bounded transfer assumption about the same identified artifacts and five contexts, not direct trace evidence on the factorial rows. It did not establish behavior on untested hardware or loads, and it did not establish cross-arm event alignment.

B. Timed-search stress test

The separate diagnostic retained the two timed-search opponents rather than silently excluding them. Each seed-condition cluster contained four profiles: serial forward, serial reverse, parallel forward, and parallel reverse. Of 200 clusters and 800 executions, 99 clusters disagreed on the frozen trace-digest endpoint (49.5%). The later digest-or-byte definition gives the same count. The 2.5th and 97.5th percentiles from the prespecified pooled whole-cluster resampling algorithm, now reported as an empirical reweighting distribution, were [42.5%, 56.5%]. This pooled procedure can change the realized context composition. The fixed counts show strong context heterogeneity: Timed-search A/orders 1 and 2 had 44/50 and 48/50 disagreements, whereas Timed-search B/orders 1 and 2 had 4/50 and 3/50. An explicitly post-acquisition, source-driven sensitivity that preserves all four stratum sizes had 2.5th and 97.5th percentiles [46.0%, 53.0%].

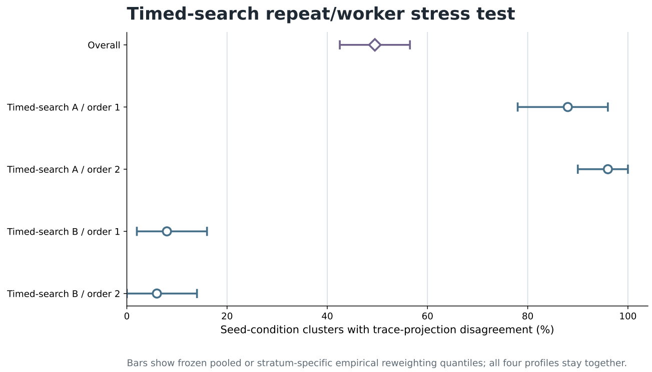

FIG. 4. Fixed-battery trace-projection disagreement in the timed-search stress test. Bars show descriptive empirical reweighting quantiles; all four execution profiles stay together within a cluster, while stratum results make the context heterogeneity visible. These quantiles are not confidence intervals. Any mismatch rejects exact parity on the exercised schedule. Neutral labels replace package names. Source data are in `source_data/figure_4_timed_search.csv`.

Outcomes disagreed in 47 clusters, decision counts in 93, and error records in 0. This rejects exact repeatability for at least one exercised seed condition and identifies plausible clock or process mechanisms; it does not prove a unique cause.

The frozen stress protocol promised first-divergence positions and acting sides plus timing summaries in the review package. A closeout search of retained evidence found acting-side counts at first divergence (all 99 disagreeing clusters localized to the opponent side) and per-profile wall-clock timing summaries in the hash-pinned stress summary; both are now included as processed aggregates with the source digest recorded, pending human approval of their redistribution. First-divergence positions were never recorded, and raw trace payloads remain restricted, so independent position-level localization cannot be verified from the release. This remains a protocol and reporting deviation for the omitted positions; it does not change the primary 99/200 digest result, but position-level, actor-side, and timing claims require that human approval before they are represented as independently reproducible, pending confirmation in the unsigned deviation record.

VIII. CLAIM-ADMISSION CONSEQUENCE

The later four-cell comparison was acquired only after the deterministic preflight passed, and its prespecified repeated-control available-record audit had 0 mismatches. Under the actual frozen rule this gate pass authorized the planned seed-matched estimates, 95% paired-resampling intervals, and secondary exact McNemar calculation for 2,000 common units per cell (12,000 engine games). The qualification preflight used a different seed range, so transfer to the same artifacts and contexts is an explicit bounded assumption. The factorial candidate rows were not trace repeated and recorded schedule, boundary seed, outcome, error, and decision count but no trace digests or semantic event identifiers. A post-acquisition methods audit therefore declines the frozen plan's inferential labels and permits only a fixed-battery descriptive seed-indexed contrast and empirical reweighting sensitivity. It admits no population effect, trace-parity, event-aligned, counterfactual, or full-CRN wording.

The total fixed-package contrast on that battery was +0.55 percentage points. The 2.5th and 97.5th percentiles from the prespecified stratified paired- unit resampling algorithm, relabeled as descriptive empirical reweighting, were [-2.05, +3.15]. They span both signs; no null test, population effect, or equivalence conclusion is drawn. Git history orders the binary gate commit before retained result artifacts, and replacing every outcome value while holding gate evidence fixed leaves that gate decision unchanged. This functional invariance does not make the later descriptive taxonomy prospective. The full factorial derivation, table, and figure are reported in the Appendix because the policy contrast is an admission demonstration rather than the protocol's principal result.

IX. INTERPRETATION AND PERMITTED CLAIMS

The evidence supports four bounded conclusions. First, matching recorded seeds does not by itself establish within-arm repeatability or cross-arm event alignment. Second, a retrospectively frozen audit of already acquired historical outcomes invalidated the planned paired decomposition; it is a rule- application example, not prospective blinding evidence. Third, the prospective deterministic preflight established repeatability for its exact artifacts, trace projection, schedule, and execution contexts, while the timed-search stress test rejected exact parity on exercised seeds. Fourth, a post-acquisition conservative report of the factorial gives fixed-battery descriptive contrasts and empirical reweighting quantiles that span both signs, without establishing a population effect, equivalence, or superiority.

The synthetic suite supports a stronger event-alignment statement only inside its logged ontology. The restricted engine cannot establish that level. Its factorial is therefore described as descriptively seed indexed and implementation conditional, not as an inferentially paired population effect or a fully coupled counterfactual comparison. The observed timed-search association does not authorize a universal claim about search agents or a unique causal attribution to wall-clock timing.

Fail-closed admission preserves useful negative results. Suppression does not delete mismatches, schedules, or diagnostics; it prevents an unsupported contrast from being promoted because it is favorable. Conversely, a passing preflight does not certify every future execution. The protocol records the scope in both directions.

X. LIMITATIONS

The conformance suite contains designed counterexamples and cannot estimate how often these failure modes occur. The restricted case study uses one engine build, fixed engineering opponent contexts, one policy-training realization, fixed seed batteries, and a declared public-observation trace projection. Opponents, seeds, and execution profiles are not probability samples, and no randomized assignment supplies a population reference distribution. The reported reweighting quantiles are therefore descriptive sensitivities for the realized batteries; they are not confidence intervals and do not generalize to a competition field, hardware population, new seed population, or new training run. The frozen plan used inferential labels for these resampling outputs; treating the same numbers only as descriptive sensitivity is a post-acquisition conservative reporting decision, not a prospectively validated claim map.

The trace projection stores hashes of public observations rather than raw observations and omits opaque search input, private simulator state, and semantic random-event identifiers. Digest–byte-count agreement is exact for the recorded projection but cannot validate omitted state; byte-count equality is a later integrity extension to the frozen digest endpoint. The preflight and factorial used different seed ranges, and factorial candidate rows contain no trace digest; their repeated-control gate compares only the frozen available record. Applying the preflight qualification to those later rows is a bounded transfer assumption rather than observed trace repeatability on the factorial battery.

The unsigned deviation record documents the frozen stress protocol's promised secondary outputs. Acting-side first-divergence counts and timing summaries were recovered from hash-pinned retained evidence and are included as processed aggregates pending redistribution approval; first-divergence positions were never recorded, so position-level localization cannot be verified from raw trace lines that remain restricted. The unrecovered positions are not treated as zero. They leave the primary digest-disagreement result intact but prevent a claim of complete protocol reporting until a human author verifies the underlying retention and approves or amends the deviation record.

The engine, third-party opponents, game assets and metadata, private replay observations, policy packages, and restricted raw traces cannot be redistributed under the authority presently established. Independent readers can regenerate the synthetic suite and reaggregate the included processed diagnostics, but cannot replay the restricted trajectories end to end. Legal ownership, a redistribution license, and an archival DOI remain unresolved and therefore block any claim that the companion package is publicly available.

Finally, author identities, affiliations, ORCIDs, CRediT roles, funding, conflicts, acknowledgments, complete historical AI-use confirmation, and permissions to identify organizations or individuals require human approval. These are submission blockers, not fields to infer from repository history.

XI. CONCLUSION

A seed schedule, a repeatable execution, and an event-aligned stochastic coupling are different scientific objects. The protocol connects each object to observable evidence and to a fail-closed reporting consequence that future users must freeze before acquisition. Its synthetic suite demonstrates the logical failure modes and an event-keyed repair; its restricted case study shows both a failed retrospective historical audit and a passing prospective qualification scope, followed by a deliberately conservative post-acquisition description of the fixed factorial battery. The practical recommendation is to identify exact artifacts and boundary seeds, audit row parity, repeat identical arms at trace-projection level across the contexts that matter, inspect stochastic sources, and suppress any claim whose required evidence is absent.

ARTIFICIAL-INTELLIGENCE ASSISTANCE

In addition to the research-conduct uses described in Methods, OpenAI Codex assisted with bibliographic verification, manuscript drafting and editing, release-package assembly, reproducibility checklists, and adversarial review. The exact deployed GPT-5-family snapshot was not exposed. Human direction and verification procedures are recorded in `supplement/AI_USE_LOG.csv`; AI suggestions were accepted only after comparison with primary sources, code, raw or processed evidence, tests, and rendered output. No AI system is an author. The corresponding author must confirm the completeness of this disclosure and compliance with applicable tool terms, confidentiality, privacy, and intellectual-property obligations.

AUTHOR CONTRIBUTIONS

Author identities and CRediT roles for conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, writing, supervision, project administration, and funding acquisition require human confirmation. No role is inferred from Git history, and OpenAI Codex is not an author.

COMPETING INTERESTS

No reliable competing-interest statement is established in the repository. Every author must disclose relevant financial and nonfinancial interests, competition affiliations, employment, sponsorship, platform access, and relationships with engine or game rights holders before submission.

DATA AVAILABILITY STATEMENT

Public (with submission, pending rights approval). The computational companion contains the synthetic protocol implementation and fixtures with expected outputs, the executable admission engine with generated decision tables (including the Stage-6 source-audit rules), and processed historical, preflight, stress, and factorial diagnostics. The processed diagnostics include recovered acting-side first-divergence counts and timing aggregates. The companion also contains analysis and verification scripts, protocol transcriptions, figures with source data, schemas, hashes, tests, environment declarations, and one-command reproduction instructions. It is computationally self-contained for the synthetic and processed analyses and does not require the game engine. Public archival availability is not yet established: ownership, an approved software/data license, archive creators, maintainer contact, and a DOI await explicit human authorization, so this manuscript does not describe the package as public or promise a later release. The unsigned chronology and omission register is included as `supplement/PROTOCOL_DEVIATIONS.md`; its pending status is neither an amendment nor human approval.

Restricted. The tournament engine and source, engine binaries, third-party opponent packages, game assets and metadata, private replay observations, policy packages, and restricted raw traces are unavailable because of organizer terms, third-party rights, privacy constraints, and unresolved release authority. They are not offered on request because the repository does not establish legal authority or a practical controlled-access mechanism; this limitation is flagged as an editorial-risk item rather than resolved by wording. Their identities are represented only by approved digests and bounded processed summaries, and exact restricted gameplay regeneration requires separately authorized access that this article cannot provide.

Appendix A: Trace record and study-specific fields

Preflight and stress executions record schedule identity, artifact digests, the value passed at the engine boundary, outcome, errors, decision count, and a canonical trace-projection digest with byte count. The projection orders each decision's public-observation hash, acting side, and chosen action, followed by the terminal record. Restricted trace records are not redistributed. Factorial records intentionally use `trace_mode=none`; they contain schedule, outcome, error, and decision fields only. This distinction prevents the factorial control gate from being described as trace parity.

Appendix B: Secondary factorial details

The factorial crossed a representation flag and a separately trained policy update, producing four fixed packages: C1 (neither), C2 (representation only), C3 (training only), and C4 (both). Each cell used the same five-context, two-order schedule with 200 paired units per stratum. Its admission gate was the prespecified repeated-C1 available-record audit described in the main text.

Let $\mu_{00}, \mu_{10}, \mu_{01}, \mu_{11}$ be the unitless win rates for C1, C2, C3, and C4, respectively, where the first subscript marks representation and the second training. The four reported percentage-point contrasts are

$$\begin{aligned}
T &= \mu_{11} - \mu_{00}, \\
R &= \tfrac{1}{2}[(\mu_{10} - \mu_{00}) + (\mu_{11} - \mu_{01})], \\
G &= \tfrac{1}{2}[(\mu_{01} - \mu_{00}) + (\mu_{11} - \mu_{10})], \\
J &= \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}.
\end{aligned} \tag{B1}$$

T is the total fixed-package intervention contrast, R the average representation contrast, G the average training contrast, and J the interaction. All are unitless before multiplication by 100. With cell rates $(0.50, 0.52, 0.51, 0.54)$, the four contrasts are $(0.04, 0.025, 0.015, 0.01)$. The signs and cell directions are tested in `tests/test_equations.py`. These algebraic contrasts do not isolate a mechanism unless the design and execution assumptions hold.

The representation, training, and interaction contrasts were +0.95, -0.40, and +1.30 percentage points, respectively; the empirical reweighting quantiles for all four contrasts span zero. This describes sensitivity to reweighting the frozen schedule-indexed units; it is not a test of a population effect. The summaries do not prove equality, equivalence, policy superiority, or event-aligned counterfactual effects.

Appendix C: Empirical reweighting details

The prespecified timed-search resampling algorithm reweights whole seed-condition clusters, retaining all four execution profiles together. Its pooled version can vary the composition of the four fixed opponent-by-order strata and has 2.5th and 97.5th percentiles [42.5%, 56.5%]. A source-driven fixed-composition sensitivity instead reweights within each 50-cluster stratum and averages the four stratum means equally; its corresponding percentiles are [46.0%, 53.0%].

TABLE IV. Fixed-battery factorial contrasts. Values are percentage-point win-rate differences with the 2.5th and 97.5th percentiles from the prespecified stratified paired-unit resampling algorithm, reported post-acquisition as empirical reweighting sensitivity rather than confidence intervals.

Contrast	Estimate	2.5th	97.5th
Total intervention	+0.55	-2.05	+3.15
Representation	+0.95	-1.08	+3.00
Training	-0.40	-2.03	+1.25
Interaction	+1.30	-1.85	+4.45

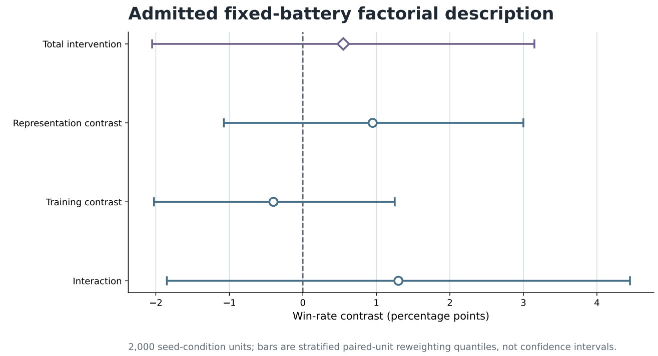

FIG. 5. Prespecified fixed-battery factorial contrasts and the 2.5th and 97.5th percentiles from the prespecified stratified paired-unit resampling algorithm, reported as empirical reweighting for 2,000 schedule-indexed units. The bars are descriptive sensitivities, not confidence intervals, equivalence tests, or statements about new opponents, seeds, or training runs. Source data are in `source_data/figure_5_factorial.csv`.

The factorial procedure reweights complete schedule-indexed paired-unit identifiers independently within each of ten opponent-by-order strata and recomputes the four cell means and contrasts. Every reported distribution uses 100,000 draws and a frozen analysis seed. Their descriptive interpretation is post-acquisition. These are empirical reweighting distributions over the observed batteries, not sampling distributions. No manuscript hypothesis test is admitted, and no equivalence margin was specified; quantiles spanning zero cannot support equivalence.

Appendix D: Artifact and rights boundary

The synthetic conformance logic, processed summaries, source-data files, protocols, and generators are separated from restricted engine and game materials. The release manifest rejects engine binaries, model weights, archives, credentials, absolute local paths, and non-allow-listed files. Dataset and artifact records state composition, intended verification use, and access restrictions rather than treating processed aggregates as replacements for restricted raw material [23].

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